

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: FPGA Based System Design (01CT0718)	Aim: To design and implement timing and event-generation systems using counters and frequency division techniques.	
Experiment No: 8	Date:	Enrolment No: 92301733013

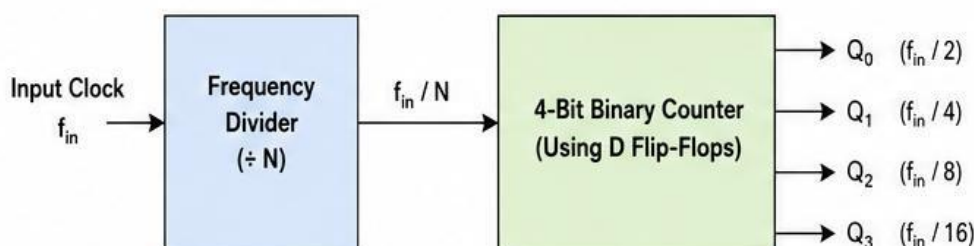
Aim: To design and implement timing and event-generation systems using counters and frequency division techniques.

Software required: Xilinx Vivado

Hardware required: Nexys4 DDR FPGA Development Board

Theory:

Block diagram:



Truth Table:

COUNT (DECIMAL)	OUTPUTS			
	Q ₃ (MSB)	Q ₂	Q ₁	Q ₀ (LSB)
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1
16 → 0	0	0	0	0 (Repeat)

Working:

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- The input clock signal f_{in} is first applied to the frequency divider.
- The frequency divider reduces the high input clock frequency by a factor of N, producing a lower-frequency clock:

$$f_{clk} = \frac{f_{in}}{N}$$

- This divided clock is applied to the 4-bit binary counter made of four D flip-flops.
- Each D flip-flop is configured to operate as a T (toggle) flip-flop by connecting the inverted output to the D input:

$$D = Q^{-}$$

Hence, the flip-flop toggles on every active clock edge.

- The first flip-flop (Q_0) toggles on every clock pulse from the frequency divider.
- Output frequency:

$$Q_0 = \frac{f_{in}}{2N}$$

- The second flip-flop (Q_1) toggles every two clock pulses.
- Output frequency:

$$Q_1 = \frac{f_{in}}{4N}$$

- The third flip-flop (Q_2) toggles every four clock pulses.
- Output frequency:

$$Q_2 = \frac{f_{in}}{8N}$$

- The fourth flip-flop (Q_3) toggles every eight clock pulses.
- Output frequency:

$$Q_3 = \frac{f_{in}}{16N}$$

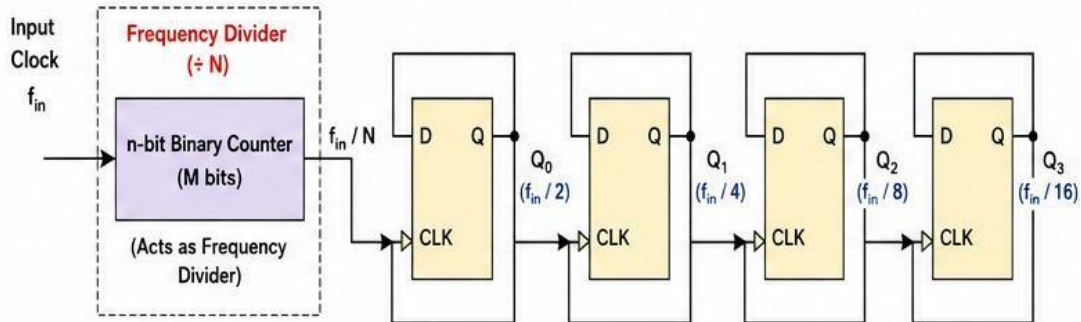
The counter counts in binary from 0000 (0) to 1111 (15). After reaching 1111, it returns to 0000 and repeats the counting sequence.

Frequency Relationship:

Signal	Frequency
Clock to Counter	f_{in}/N
Q_0	$f_{in}/2N$
Q_1	$f_{in}/4N$
Q_2	$f_{in}/8N$
Q_3	$f_{in}/16N$

Circuit Diagram:

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Verilog Code:

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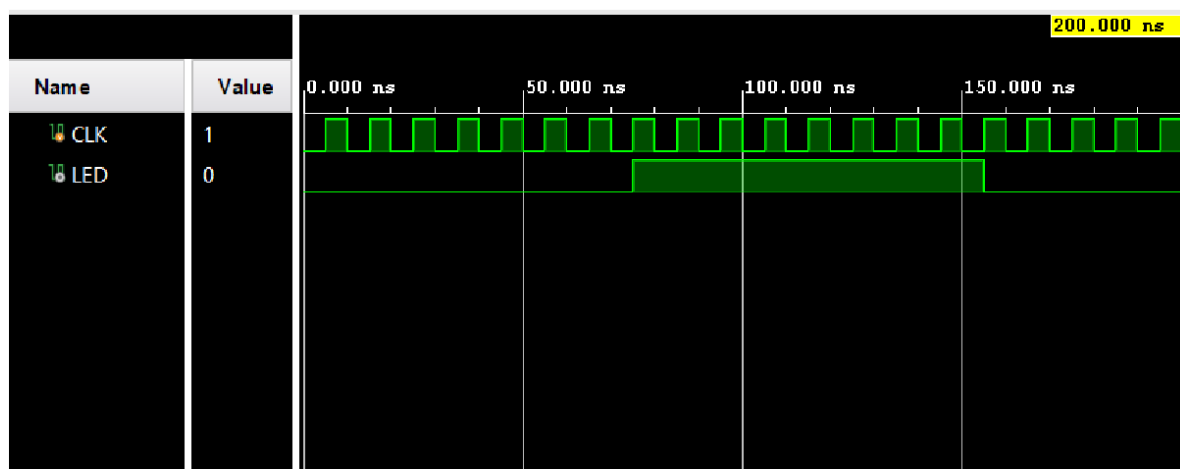
module counter_divider(
    input CLK,
    output LED
);
    reg [25:0] counter = 26'd0;

    always @(posedge CLK)
    begin
        counter <= counter + 1;
    end

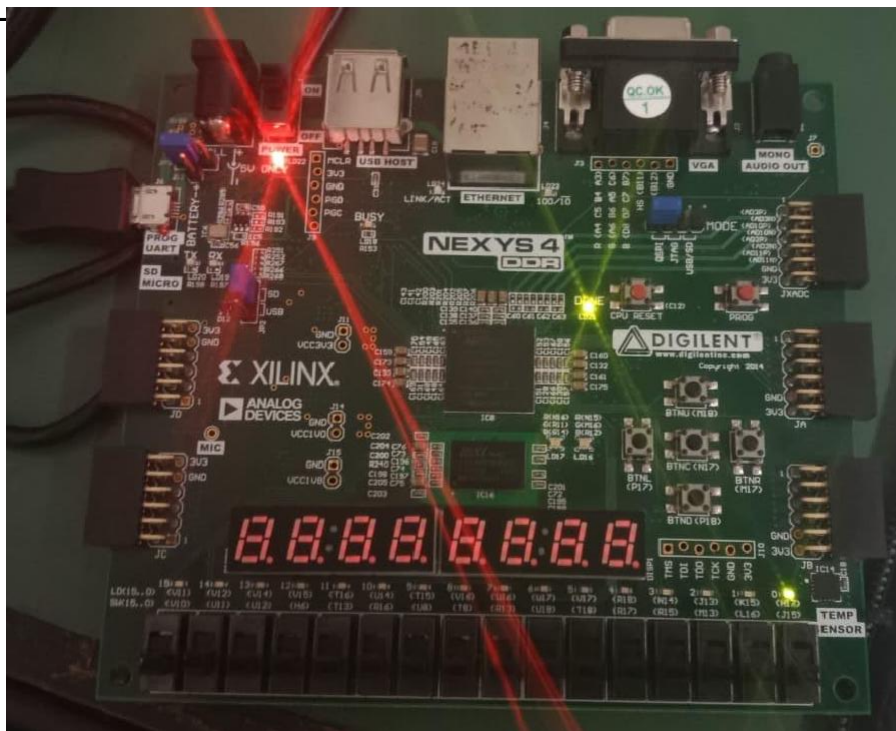
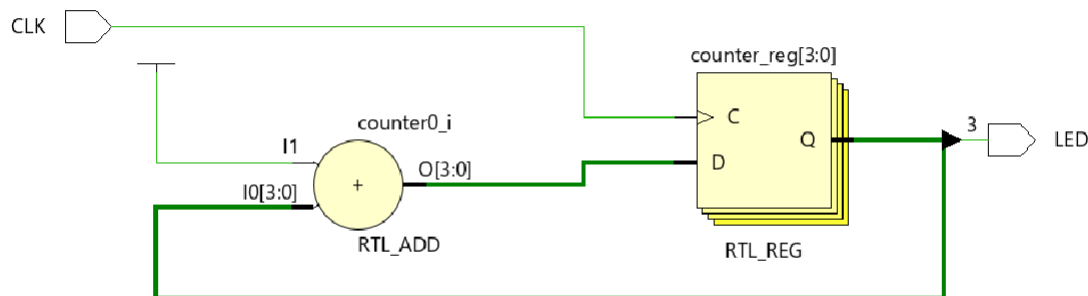
    assign LED = counter[25];
endmodule

```

Simulation Results:



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Conclusion:

The 4-bit counter with a frequency divider was successfully designed and implemented on an FPGA using D flip-flops. The frequency divider lowered the high-speed input clock to a suitable frequency, allowing the counter to operate correctly and count from 0000 to 1111. Both simulation and hardware testing confirmed the expected frequency division and binary counting behavior. This experiment demonstrated the practical application of counters in digital timing and event-generation systems.

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Post Lab Exercise:

1. Why is a frequency divider required before driving a counter on an FPGA?

A frequency divider converts the high-frequency FPGA system clock, such as 50 MHz or 100 MHz, into a lower frequency that is suitable for the counter. This allows the counter to run at a visible and manageable speed instead of counting too quickly. It also helps maintain proper timing in applications such as digital clocks, LED displays, and timers.

2. How can power consumption be reduced in FPGA-based counter circuits?

- Using **clock enable (CE)** instead of continuously toggling the counter.
- Reducing the operating clock frequency using a frequency divider.
- Disabling unused logic blocks.
- Using synchronous designs with dedicated clock networks to minimize unnecessary switching activity.

3. What is the advantage of using dedicated clock resources for frequency divider circuits?

- Low clock skew
- Lower propagation delay.
- Reliable and stable clock distribution.
- Better timing performance, allowing the circuit to operate at higher speeds without timing violations.

4. What is the impact of propagation delay on high-speed counter operation?

- Incorrect counting.
- Timing errors.
- Setup and hold time violations.
- Reduced maximum operating frequency of the counter.

5. How does the division factor of a frequency divider affect the counter output?

Higher division factor → Lower clock frequency → Counter increments more slowly.

Lower division factor → Higher clock frequency → Counter increments more quickly.